Encyclopedia entry

Makeism

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Makeism

Makeism is a methodological position in cognitive science and artificial intelligence that stipulates that genuine understanding of cognition is achieved by constructing it. According to this view, if cognition is computational, then it should, in principle, be possible to recreate cognitive capacities in artificial systems. Successful construction is therefore treated not just as imitation of cognitive processes, but as explanation. Although historically influential, Makeism has recently been criticized as conceptually and computationally untenable (van Rooij et al., 2024).

Core Claims

Makeism can be understood as a strong interpretation of computationalism, rooted in the physical symbol systems hypothesis (Newell, 1980). From this starting point, van Rooij et al. (2024) characterize Makeism as advancing three interdependent claims.

First, practical feasibility proposes that if cognition, or a substantive cognitive capacity, is computational in nature, it should be replicable in artificial systems. Building on this, explanatory sufficiency holds that successfully constructing cognition is sufficient for explaining it. In other words, if a system behaves like a mind, then the mechanisms used to build it reveal how cognition itself works.

The strongest form of Makeism asserts explanatory necessity, which argues that reconstruction of cognition is requirement for genuine explanation and understanding. Within this framework, cognition cannot be understood without first being built. Logically, if practical feasibility fails, then the subsequent claims collapse.

The concept is well summarized by Richard Feynman’s dictum, “What I cannot create, I do not understand.”

Historical Context

Makeism played a central role in early cognitive science and AI. Researchers such as Herbert Simon argued that general information-processing devices, specifically digital computers, could implement any cognitive process, in principle (Simon, 1983). This assumption motivated synthetic methodologies in which cognitive theories were implemented as executable programs with the idea that if intelligent behavior was reproduced, there was evidence that the theory explained the mind.

These methodologies made Makeist approaches vulnerable to a form of map-territory confusion, in which computational models are mistaken for the real cognitive processes they are meant to describe and are treated as if they directly revealed how human cognition actually works.

Critiques

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Fundamental objections to Makeism have been raised in recent work, challenging the main assumption that cognition can be feasibly constructed in artificial systems. Van Rooij et al. (2024) show that realistic cognitive systems would require solving problems which are far too complex for any presently achievable system. As a result, systems produced by Makeist approaches are likely to remain approximations instead of genuine models of human cognition.

This criticism aligns with Marr's (1982), stating that construction does not, by itself, constitute explanation because explanation requires analysis at the computational, algorithmic, and implementation levels.

Importantly, rejecting Makeism does not entail abandoning computational modeling altogether. In fact, Makeism marks a foundational moment in the history of cognitive science by tying explanation to construction. Its recent critique has reshaped how the field understands what computational models can achieve, with them now valued primarily as instruments for theoretical insight.

References

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